

CENG381 Stochastic Processes -- Mixed Exam

Mixed Exam 4 -- Renewal-Reward | Birth-Death & Queues | Wald & De Moivre

Duration: 90 minutes. Total: 100 points. Show all working.

Notation: $C(t)$ = total reward by time t . $B(t)$ = residual life at time t . $r = q/p$ = De Moivre ratio. T = random stopping time with $E[T] < \infty$.

Q1 (30 points). A city bus runs on a loop. Each round-trip has two phases: Outbound (O) with random duration $U(O) \sim \text{Uniform}[20, 40]$ minutes, and Return (R) with duration $U(R) \sim \text{Exponential}$ with mean 25 minutes. Each trip earns a revenue of $R(n) = 50 * U_O(n)$ dollars (proportional to outbound time). Cycles are i.i.d. Define the renewal point as the start of each outbound trip.

a) Compute $E[X]$ (mean cycle length) and $E[R]$ (mean revenue per cycle). For $U(O) \sim \text{Uniform}[20, 40]$: $E[U(O)] = 30$ and $E[U(R)] = 25$.

b) Apply the Renewal-Reward Theorem to find the long-run revenue rate (dollars per minute):

$$\text{Long-run rate} = E[R] / E[X]$$

How many dollars per hour does the bus earn on average in the long run?

c) Compute $E[X^2]$ for the cycle length. Use $\text{Var}[U(O)] = (40-20)^2/12$ for uniform, $\text{Var}[U(R)] = 25^2$ for exponential, and $E[X^2] = \text{Var}[X] + (E[X])^2$. Assume $U(O)$ and $U(R)$ are independent.

Then compute $E[B] = E[X^2]/(2E[X])$, the expected residual time until the next departure for a random passenger arrival.

Q2 (35 points). A hospital ward has n beds. Patients arrive at rate $\lambda = 6$ per hour and each occupies a bed for an Exponential duration with mean $1/\mu = 0.5$ hours. The ward has INFINITE capacity (M/M/ ∞ model).

a) State the M/M/ ∞ stationary distribution $\pi(n)$ and identify it as a named distribution with parameter $\rho = \lambda/\mu$. Compute ρ and find $P(\text{ward has 0 patients})$ and $P(\text{ward has exactly 3 patients})$.

b) Find $E[N]$ (mean occupancy) and $E[T]$ (mean time each patient spends). Compare $E[T]$ to $1/\mu$: are patients spending time waiting, or just being treated?

c) The hospital considers switching to a single-doctor model (M/M/1) where one doctor treats patients at rate $\mu' = \lambda + 1 = 7$ per hour (just above the arrival rate to ensure stability). Compute ρ' , $E[N']$ and $E[T']$ for this M/M/1 queue. Which model gives shorter $E[T]$: M/M/ ∞ with $\mu=2$ or M/M/1 with $\mu'=7$?

CENG381 Stochastic Processes -- Mixed Exam

Mixed Exam 4 -- Renewal-Reward | Birth-Death & Queues | Wald & De Moivre

Q3 (35 points). A gambler plays a biased coin: heads (+1) with probability $p = 2/3$, tails (-1) with probability $q = 1/3$. Starting fortune: $k = 4$. Absorbing barriers at 0 (ruin) and $N = 10$ (goal).

a) Compute $r = q/p$. Write De Moivre's martingale $M(n) = r^{S(n)}$. Verify it is a martingale by showing $E[r^{X(n+1)}] = 1$.

b) Apply OST to $M(n)$ with stopping time $T = \text{first hit of } \{0, 10\}$. Let $h = P(\text{reach 10 before 0} \mid \text{start } k = 4)$. Set up and solve:

$$r^{10} \cdot h + 1 \cdot (1 - h) = r^4$$

c) The gambler's total winnings at stopping is $S(T)$. Using Wald's equality $E[S(T)] = E[X] \cdot E[T]$ where $E[X] = p - q$, express $E[T]$ in terms of h, k, N, p, q , and $E[S(T)]$.
Compute $E[S(T)] = 10 \cdot h + 0 \cdot (1 - h) = 10h$, then find $E[T]$.